

Date	06 July 2026	Project Name	Heart Disease Analysis
Team ID	—	Maximum Marks	4 Marks

IDEATION PHASE

Brainstorm & Idea Prioritization Template

Brainstorming provided a free and open environment that encouraged every member of the team to participate in the creative-thinking process leading to problem solving for the project, **Heart Disease Analysis**. Prioritising volume over value, out-of-the-box ideas were welcomed and built upon, and all participants collaborated to develop a rich set of analytical and design ideas.

This document records how the team used a structured brainstorming and prioritization process to move from an open problem space to a focused, feasible project plan, even while coordinating across individual schedules.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team **Mounika Sanapathi (Team Lead)**, **Thakasi Balaji**, **Kale Sai Praneeth**, **Navyasri Gonnuri**, and **Shanthi Kumari Challa** came together to define the scope of the project. Relevant background on clinical and cardiovascular datasets and prior Tableau dashboard examples was shared with all members ahead of the session, and each member confirmed their availability and role before the session began.

PROBLEM STATEMENT

"How might we analyse cardiovascular patient clinical metrics and identify key risk factors using data analytics, so that healthcare providers and medical professionals can make informed, data-driven decisions about patient care, diagnosis prioritization, and preventive health strategies?"

Key Rules of Brainstorming Followed

- ✓ **Stay on topic:** Keep every idea tied to the problem statement on heart disease risk factors and patient clinical metrics.
- ✓ **Encourage wild ideas:** Unconventional analytical angles (such as complex metric correlations) were welcomed before being filtered.
- ☑ **Defer judgment:** No idea was rejected during the listing stage; evaluation came later, in Step-3.
- ✓ **Listen to others:** Each member reviewed and built upon ideas shared by teammates.
- ✓ **Go for volume:** The goal in Step-2 was quantity of ideas, not immediate quality.
- ✓ **Be visual:** Ideas were grouped and plotted visually to aid shared understanding.

Step-2: Brainstorm, Idea Listing and Grouping

Individual Idea Listing (Sticky-Note Stage)

Each member independently listed ideas addressing the problem statement, without judgment or discussion, before the ideas were shared and clustered as a team.

Mounika Sanapathi (Lead)	Thakasi Balaji	Kale Sai Praneeth	Navyasri Gonnuri	Shanthi Kumari Challa
Collect heart disease patient data covering age, sex, cholesterol levels, blood pressure, Max HR, and presence of disease.	Study correlation between physiological metrics across different age groups to find risk patterns.	Design an interactive Tableau dashboard showing disease prevalence by age group, sex, and chest pain type.	Build a responsive website to host and present all project medical insights and outputs.	Add dashboard filters for age bracket, sex, blood pressure range, and fasting blood sugar levels.
Standardise and clean raw clinical data to remove duplicate, missing, or inconsistent medical records.	Segment patients by risk categories: low-risk, moderate-risk, and high-risk groups.	Build a Tableau Story to narrate key clinical insights and diagnostic findings in a logical sequence.	Publish the Tableau dashboard on Tableau Public for open healthcare research accessibility.	Use distribution charts and correlation heat maps to visualise patient risk density by metric.
Include KPI cards for total patients analysed, average cholesterol levels, and percentage of high-risk cases.	Analyse the impact of resting ECG results and exercise-induced angina on final diagnoses.	Track average cholesterol and blood pressure trends across different patient demographic segments.	Host project code, clinical datasets, and documentation on GitHub for transparency.	Ensure the website and embedded clinical dashboard are mobile-friendly for doctors on the move.
Explore the correlation between maximum heart rate achieved during exercise and heart disease severity.	Investigate patient correlation patterns across multiple clinical parameters (e.g., ST depression, slope).	Add a summary/ insights page on the website linking to the live dashboard and clinical story.		

Grouped Ideas (Thematic Clusters)

Once individual ideas were shared, similar and related ideas were clustered into themes, and each cluster was given a short label reflecting its focus area.

Cluster A: Data Collection & Preparation (Led by: Mounika Sanapathi)

- Sourcing heart disease data (age, sex, cholesterol, blood pressure, max heart rate, disease presence).
- Cleaning and standardising raw clinical data, removing duplicates and missing medical values.

- Identifying clinical risk patterns and data trends across patient demographics.

Cluster B: Patient Behaviour & Risk Segmentation (Led by: Thakasi Balaji & Kale Sai Praneeth)

- Segmenting patients by cardiovascular risk levels and diagnostic categories.
- Studying the effect of resting ECG, chest pain types, and exercise-induced metrics on diagnoses.
- Analysing correlations between blood pressure, cholesterol, and final heart disease presence.

Cluster C: Dashboard & Visualization Design (Led by: Shanthi Kumari Challa)

- Building the interactive Tableau dashboard with patient filters and clinical KPI cards.
- Creating the Tableau Story to communicate medical insights sequentially.
- Designing correlation maps and risk density charts by demographic and clinical parameters.

Cluster D: Deployment & Accessibility (Led by: Navyasri Gonnuri)

- Publishing the clinical dashboard on Tableau Public.
- Building and hosting the responsive health project website.
- Maintaining the GitHub repository for medical analytics code and dataset documentation.

Step-3: Idea Prioritization

With the clustered ideas in hand, the team plotted each idea on an Importance-versus-Feasibility grid to agree on what mattered most and what was realistically achievable within the project timeline and available clinical dataset.

Idea Prioritization Grid — Importance vs. Feasibility

- **DO FIRST:** Data Cleaning & Preparation; Interactive Tableau Dashboard; Clinical Metric & Risk Factor Analysis.
- **PLAN NEXT:** Patient Risk Segmentation Analysis; Predictive Diagnostic Trend Modelling; ECG Correlation Impact Analysis.
- **FILL-IN TASKS:** Tableau Story Narrative; Responsive Website & Public Deployment; Demographic Preference Study.
- **RECONSIDER:** Advanced Multi-metric Chemical Analysis (retained only as a secondary insight, not a core deliverable).

Prioritization Summary

Quadrant	Ideas
Do First	Data Cleaning & Preparation; Interactive Tableau Dashboard; Patient Risk Segmentation Analysis
Plan Next	Predictive Diagnostic Trend Modelling (kept as future scope due to limited feasibility with the current cross-sectional dataset)

Quadrant	Ideas
Fill-in Tasks	Tableau Story Narrative; Responsive Website & Public Deployment; Patient Demographic Study
Reconsider	Advanced Multi-metric Chemical Analysis (retained only as a secondary insight, not a core deliverable)

The ideas placed in the "Do First" quadrant—data cleaning and preparation, the interactive Tableau dashboard, and patient risk segmentation analysis—formed the core of the completed project, and were carried forward into the dataset collection, cleaning, Tableau Dashboard, Tableau Story, and Tableau Public deployment stages.

The Tableau Story narrative and the responsive website/public deployment were treated as important supporting deliverables and were completed alongside the core health analysis.

Predictive diagnostic trend modelling was consciously deferred as future scope, since it required a longer, longitudinal clinical time-series dataset than was feasible for the current project scope.